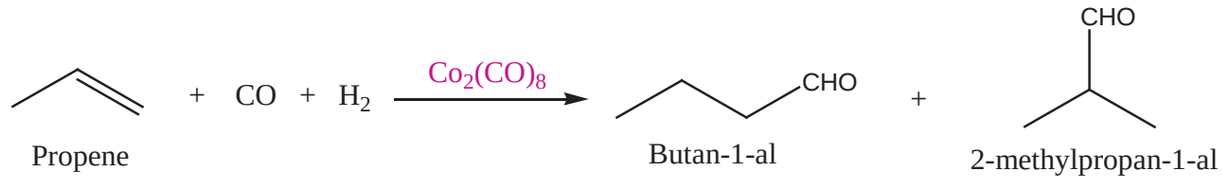
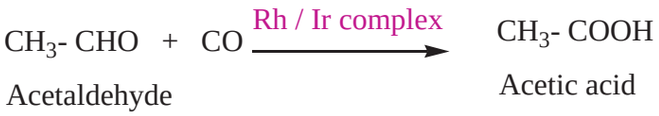
(V_2O_5) is used as a catalyst to oxidise SO_2. In this reaction V_2O_5 is reduced to vanadium (IV) Oxide (VO_2).

Some more examples are discussed below,

(i) Hydroformylation of olefins

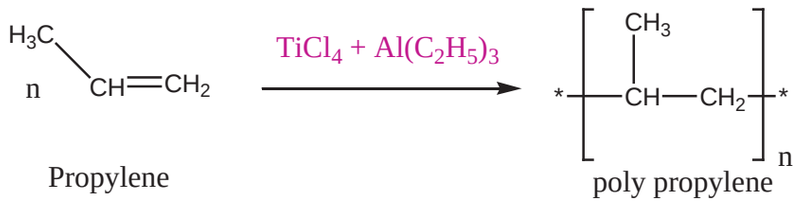


(ii) Preparation acetic acid from acetaldehyde.



(iii) Zeigler – Natta catalyst

A mixture of TiCl4 and trialkyl aluminium is used for polymerization.



4.3.8 Alloy formation

An alloy is formed by blending a metal with one or more other elements. The elements may be metals or non-metals or both. The bulk metal is named as solvent, and the other elements in smaller portions are called solute. According to Hume-Rothery rule to form a substitute alloy the difference between the atomic radii of solvent and solute is less than 15%. Both the solvent and solute must have the same crystal structure and valence and their electro negativity difference must be close to zero. Transition metals satisfying these mentioned conditions form a number of alloys among themselves, since their atomic sizes are similar and one metal atom can be easily replaced by another metal atom from its crystal lattice to form an alloy. The alloys so formed are hard and often have high melting points. Examples: Ferrous alloys, gold – copper alloy, chrome alloys etc.,

4.3.9 Formation of interstitial compounds

An interstitial compound or alloy is a compound that is formed when small atoms like hydrogen, boron, carbon or nitrogen are trapped in the interstitial holes in a metal lattice. They are usually non-stoichiometric compounds.Transition metals form a number of interstitial compounds such as TiC, $ZrH_{1.92}$, Mn_4N etc. The elements that occupy the metal lattice provide them new properties.

(i) They are hard and show electrical land thermal conductivity

(ii) They have high melting points higher than those of pure metals

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